

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Implementation of Basic Student Profile using JavaScript Classes and DOM

2. AIM

To construct an interactive student profile generator using JavaScript ES6 classes to encapsulate student data, and DOM manipulation techniques to dynamically render the data on the webpage.

3. OBJECTIVE

The objectives of this experiment are:

- To define JavaScript ES6 classes with constructors for structuring data.
 - To instantiate objects from defined classes using user-provided input.
 - To manipulate the Document Object Model (DOM) to capture user input values.
 - To dynamically inject HTML content and display structured object properties on the webpage.
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4. THEORY

JavaScript ES6 introduced the `class` keyword, providing a much simpler and clearer syntax to create objects and deal with inheritance. A class acts as a blueprint for creating objects, containing a `constructor` method to initialize properties.

The Document Object Model (DOM) connects web pages to scripts. Methods like `getElementById` allow developers to fetch user input, while the `innerHTML` property allows dynamic insertion of HTML nodes into the document to display output.

5. ALGORITHM / PROCEDURE

1. Create an HTML layout with input fields for Name, Roll No, Department, and CGPA.
 2. Add a styled button configured to trigger a JavaScript function `showProfile()` on click.
 3. Define an ES6 class `Student` with a constructor to initialize properties.
 4. In the function, retrieve the values from the input fields using DOM methods.
 5. Implement validation to alert the user if any fields are empty.
 6. Instantiate a new `Student` object passing the input values.
 7. Use template literals to construct HTML strings mapping the object's properties.
 8. Update the `innerHTML` of the profile display container to show the student details.
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6. PROGRAM

Source File: task2.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Profile Generator</title>
  <link rel="stylesheet" href="task2.css">
</head>
<body>

  <h1>Student Profile Generator</h1>

  <!-- User Input Form -->
  <div class="form-container">
    <div class="input-group">
      <label for="name">Name:</label>
      <input type="text" id="name" placeholder="">
    </div>

    <div class="input-group">
      <label for="rollNo">Roll No:</label>
      <input type="text" id="rollNo" placeholder="">
    </div>

    <div class="input-group">
      <label for="department">Department:</label>
      <input type="text" id="department" placeholder="">
    </div>

    <div class="input-group">
      <label for="cgpa">CGPA:</label>
      <input type="number" step="0.01" id="cgpa" placeholder="">
    </div>

    <button id="generateBtn">Generate Profile</button>
  </div>

  <!-- Container for dynamically generated profile -->
  <div id="profileContainer"></div>

  <script src="task2.js"></script>
</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
class Student {}	Defines a JavaScript ES6 class acting as an object blueprint.

constructor(...)	Special method for creating and initializing an object created with a class.
document.getElementById()	DOM method that returns the element with the specified ID.
element.value	Property that sets or returns the value of an input field.
new Student(...)	Instantiates a new object instance of the Student class.
element.innerHTML	Sets or returns the HTML content (inner markup) of an element.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Empty Input	Click button with empty fields	Triggers alert warning for empty fields	Pass
TC-02	Object Init	Enter details & submit	Instantiates Student class successfully	Pass
TC-03	DOM Injection	Verify profile output section	HTML dynamically displays entered values	Pass

9. OUTPUT

Output 1

The screenshot displays a web browser window with the title 'Student Profile Generator'. The address bar shows the URL `127.0.0.1:5500/observation/obs_task2/source_code/task2.html`. The form contains the following elements:

- Name:**
- Roll No:**
- Department:**
- CGPA:**
- Generate Profile** (button)

The Windows taskbar at the bottom shows the date and time as 8/31/26, 10:43 PM, the weather as 30°C Mostly cloudy, and the search bar.

10. RESULT

Thus, the basic student profile generator utilizing JavaScript ES6 classes and DOM manipulation was successfully implemented and verified.